

Elias Dunham

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EDUCATION

Northeastern University, College of Engineering

Candidate for Bachelor of Science in Mechanical Engineering and Physics

Boston, MA

May 2028

GPA: 3.8

Coursework: Fluid Mechanics, Thermodynamics, Mechanics of Materials, Materials Science, Differential Equations, Physics 2

Extracurriculars: American Society of Mechanical Engineers, Northeastern Composition Club, Delta Tau Delta

Northeastern University, Dialogue of Civilizations

International Applications of Fluid Mechanics

Panama City, Panama

July 2026

Studied fluid mechanics via case studies at the Panama Canal, a Penonomé wind farm, and Chiriquí hydroelectric facilities

Institut Notre-Dame de Bonne-Espérance

Rotary Youth Exchange Program

Braine-le-Comte, Belgium

June 2024

Study abroad focused on cultural immersion and French language

Southern Lehigh High School

Extracurriculars: Class Vice President, Future Business Leaders of America President, Tennis Team Captain

Center Valley, PA

May 2023

PROFESSIONAL EXPERIENCE

MIT Lab for Translational Engineering

Mechanical Engineering Co-op

Cambridge, MA

January - June 2026

- Owned mechanical design and fabrication across multiple concurrent device programs, taking each from CAD through prototyping and in-vivo validation
- Delivered experiment-ready hardware for postdoctoral researchers, producing over 40 assembled devices and 320 printed and machined parts to specification
- Established repeatable fabrication and test protocols, training lab members to run them independently

FLX Solutions

Mechanical Engineering Intern

Bethlehem, PA

June - August 2025

- Collaborated with team to duplicate company's testbed for evaluating embedded software on hardware
- Programmed a custom application with JavaScript and OpenAI to fully automate company's task assignment protocol
- Created and trained a custom LLM using Python and Microsoft TrOCR to automatically integrate handwritten notes into the company's information database

Projects

Wetsuit Thermal Research | US Navy, MIT Lab for Translational Engineering

March 2026 - Present

- Engineered a high-pressure testing apparatus with PID-controlled hot plate to evaluate the thermal conductivity of experimental diving materials at depth
- Conducted mechanical stress testing on novel materials to characterize structural integrity under high-pressure cycles

Stem-Cell Delivery Device | MIT Lab for Translational Engineering

February 2026 - Present

- Designed a precision delivery actuator for regenerative medicine applications used for *in-vivo* testing
- Engineered a low-friction geometry to maintain laminar flow, preventing cell shearing and ensuring high percentage of cell viability

Vagus Nerve Stimulation PCB Housing | MIT Lab for Translational Engineering

February - April 2026

- Designed and fabricated a specialized PCB housing for vagus nerve stimulator to protect sensitive electronics during *in-vivo* trials
- Optimized housing geometry and sealing to meet the rigorous structural and biocompatibility requirements

Surgical Tool Sanitization System | MIT Lab for Translational Engineering

January - February 2026

- Engineered a high-pressure rinsing system for surgical instruments using spring-loaded nozzle and capillary-action fluid restriction
- Developed a manual, spring-actuated jet mechanism to deliver high-velocity fluid for the removal of biological debris from tools

Biomimetic Fan Blade Technology | Cornerstone of Engineering

January - March 2025

- Used SolidWorks to develop fan blades based on the mechanics of owl wings for applications in drone manufacturing and HVAC
- Blades proven to minimize turbulence and noise generation by 30% compared to industry standards

COMPUTER & MACHINING SKILLS

CAD: SolidWorks, Autodesk Fusion, AutoCAD

Programming: C++, JavaScript, Python, MATLAB, Arduino

Manufacturing: 3D Printing (SLA, FDM, PµSL), CNC (3 & 5-axis mill), Manual Injection Molding, Laser Cutting

Testing: Mechanical Testing (Instron, ADMET), Hydraulic Press, Plasma Cleaner, Vacuum Compression Molding

Electronics: PID Control Systems, Oscilloscopes, Soldering, Circuit Engraving (Laser Ablation)

AWARDS & INTERESTS

Awards: Dean's List, Commended National Merit Scholar, Second Place Future Business Leaders of America Pennsylvania State Finals

Interests: Classical Piano Performance, Astronomy, Golf, Cross-Country Skiing